

Code No: 155AY**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD****B. Tech III Year I Semester Examinations, August - 2022****DISTRIBUTED DATABASES
(Computer Science and Engineering)****Time: 3 Hours****Max. Marks: 75****Answer any five questions
All questions carry equal marks**

- 1.a) Define Distributed DBMS? Discuss problem areas in DDBS environment.
- b) What is transparency in DDBMS? Explain Layers of transparency in DDBMS. [8+7]
- 2.a) Explain Client Server architecture for Distributed DBMS.
- b) What is allocation? List and explain the information requirements during allocation. [8+7]
- 3.a) Draw and explain query processing in centralized system.
- b) Write a short note on Phases of Distributed Query Processing. [8+7]
- 4.a) Explain objectives of Query Processing.
- b) Discuss Query Optimization in brief. [6+9]
- 5.a) Explain View management. How it is carried out in centralized and distributed DBMS?
- b) Define a transaction. Discuss the ACID properties of a transaction. [7+8]
- 6.a) Compare locking-based and timestamp-based concurrency control protocols with the help of examples.
- b) Explain serializable schedules with the help of example. [8+7]
- 7.a) What are parallel execution problems? Explain.
- b) Explain the reliability concept and majors. [8+7]
- 8.a) Write short notes on object migration.
- b) Describe in detail cache consistency and object identifier management in object management. [6+9]

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Code No: 155AY**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD****B. Tech III Year I Semester Examinations, February - 2022****DISTRIBUTED DATABASES****(Computer Science and Engineering)****Time: 3 hours****Max. Marks: 75**

Answer any five questions
All questions carry equal marks

- 1.a) Explain the various problem areas in DDBS environment.
b) Discuss the distributed database design issues. [8+7]
- 2.a) Explain in brief about vertical fragmentation.
b) Explain in brief architectural models for distributed DBMS. [7+8]
- 3.a) Discuss the problems in query optimization in DDBs.
b) Discuss the objectives of query processing. [8+7]
4. Explain any two distributed query optimization algorithms in brief. [15]
- 5.a) Explain various types of transactions.
b) Explain the concept of serializability. [7+8]
6. Explain in detail about time-stamp based concurrency control algorithm. [15]
- 7.a) Discuss the failures in distributed DBMS.
b) Explain distributed reliability protocols. [7+8]
- 8.a) Discuss the architectural issues in distributed object DBMS.
b) Explain in brief about distributed object storage. [8+7]

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Code No: 155AY**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD****B. Tech III Year I Semester Examinations, January/February - 2023****DISTRIBUTED DATABASES****(Common to CSE, CSE(N))****Time: 3 Hours****Max. Marks: 75****Note:** i) Question paper consists of Part A, Part B.

ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.

iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A**(25 Marks)**

- 1.a) What are the properties of data processing? [2]
- b) Write promises of DDBSs. [3]
- c) What is query translation? [2]
- d) Give the importance of localization of distributed data. [3]
- e) List the types for deadlock avoidance in distributed system. [2]
- f) Define serializability with examples. [3]
- g) Write the advantages of parallel distributed database. [2]
- h) Write short notes on fault-tolerance in distributed systems. [3]
- i) Define inheritance. [2]
- j) Write architectural issues of object database. [3]

PART – B**(50 Marks)**

- 2.a) Write and explain design strategies of distributed database design.
- b) Why we need to have distributed databases, and distinguish the features of distributed databases with centralized databases. [5+5]

OR

- 3.a) Draw and explain distributed database management system architecture.
 - b) Illustrate allocation methods for distributed database design. [5+5]
- 4.a) Discuss in detail about distributed query optimization algorithms.
 - b) How is distributed grouping and aggregate function evaluation done? [5+5]

OR

- 5.a) Discuss in detail about objectives of Query Processing.
 - b) Describe characterization of query processors. [5+5]
- 6.a) Differentiate between centralized 2PL protocol and Distributed 2PL protocol.
 - b) List and explain types of transactions with suitable diagrams. [5+5]

OR

- 7.a) Elaborate Distributed Deadlock detection for distributed database systems.
- b) Explain optimistic concurrency control algorithms. [5+5]

- 8.a) List and discuss various types of failures in distributed DBMS.
b) Discuss in detail about reliability concept and majors. [5+5]

OR

- 9.a) Differentiate between local and distributed reliability protocol.
b) Explain general architecture of a parallel database system and shared memory architecture. [5+5]

- 10.a) Describe in detail cache consistency and object identifier management in object management.
b) Discuss architectural issues in distributed object DBMS. [5+5]

OR

- 11.a) Compare QODBMS and ORDBMS.
b) Discuss in detail about persistent programming languages. [5+5]

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Code No: 155AY**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD****B. Tech III Year I Semester Examinations, July/August - 2023****DISTRIBUTED DATABASES****(Common to CSE, CSE(N))****Time: 3 Hours****Max. Marks: 75**

- Note:** i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A**(25 Marks)**

- 1.a) What are the distribution design issues? [2]
- b) What are the information requirements during allocation? [3]
- c) What are the layers of query processing? [2]
- d) What is query optimization? [3]
- e) What are the types of Transaction? [2]
- f) What is meant by deadlock? List out the conditions for deadlock? [3]
- g) What is meant by network partitioning? [2]
- h) What are the reliability protocols in distributed systems? [3]
- i) What are the applications of object identification? [2]
- j) What is meant by distributed object storage? [3]

PART – B**(50 Marks)**

- 2.a) Explain the principles of distributed databases.
 - b) Explain about Architectural Models for DDBMS. [5+5]
- OR**
- 3.a) Explain Briefly about DDBMS Architecture.
 - b) Briefly explain distributed data processing. [5+5]
- 4.a) Explain about query processing problems.
 - b) Explain briefly about query decomposition and data localization. [5+5]
- OR**
- 5.a) Briefly describe the characterization of query processors.
 - b) Explain any one distributed query optimization algorithm. [5+5]
- 6.a) Briefly explain transaction management.
 - b) Explain deadlock management. [5+5]
- OR**
- 7.a) Explain different optimistic methods for distributed concurrency control.
 - b) Explain the distributed deadlock detection. [5+5]

- 8.a) Explain the failures in DDBMS.
b) What are different types of Parallel databases architecture? [5+5]

OR

- 9.a) Briefly explain fault-tolerance in distributed systems.
b) What is parallel data placement? Explain with an example? [5+5]
- 10.a) Explain Fundamental object concepts with an example.
b) Briefly explain persistence of objects. [5+5]

OR

- 11.a) Comparison between OODBMS and ORDBMS.
b) Explain object query processing with example. [5+5]

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Code No: 155AY**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD****B. Tech III Year I Semester Examinations, March - 2024****DISTRIBUTED DATABASES**
(Computer Science and Engineering)**Time: 3 Hours****Max. Marks: 75****Note:** i) Question paper consists of Part A, Part B.

ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.

iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A**(25 Marks)**

- 1.a) Define a distributed database system. [2]
- b) List the key issues in distribution design. [3]
- c) Name any two objectives of query optimization. [2]
- d) Write a short note on query localization. [3]
- e) What is the significance of a commit point in a transaction? [2]
- f) Differentiate between centralized and distributed transactions. [3]
- g) Define database clusters. [2]
- h) How do distributed reliability protocols handle failures? [3]
- i) What is mean by distributed object storage? [2]
- j) Name a few persistent programming languages commonly used in object-oriented data modeling. [3]

PART – B**(50 Marks)**

- 2.a) Discuss the significance of distributed data processing in modern computing.
 - b) Elaborate on the promises made by distributed database systems and their impact. [5+5]
- OR**
- 3.a) Explain the concept of fragmentation and its role in distributed database design.
 - b) Write a detailed overview of the components in distributed database management system architecture. [5+5]
- 4.a) Discuss the main objectives of query processing and how they contribute to the efficiency of database systems.
 - b) Compare and contrast centralized and distributed query optimization. [5+5]
- OR**
5. Describe the layers of query processing in detail, highlighting the functions of each layer. [10]
 6. Explain the types of serializability with suitable examples. [10]
- OR**
- 7.a) Describe the types of transactions and provide examples of each.
 - b) Discuss two-phase locking as a concurrency control mechanism. [5+5]

8. Explain the process of parallel query processing. Illustrate with an example. [10]

OR

9. Analyze the impact of site failures and network partitioning on the reliability of Distributed DBMS. [10]

10.a) Discuss the fundamental object concepts and models in DODBMS.

b) Explain the characteristics of persistent programming languages in the context of object-oriented data modeling. [6+4]

OR

11.a) Compare OODBMS with ORDBMS.

b) Elaborate on the concept of object identity. [5+5]

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Code No: 155AY**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD****B. Tech III Year I Semester Examinations, August/September - 2024****DISTRIBUTED DATABASES****(Computer Science and Engineering)****Time: 3 Hours****Max. Marks: 75****Note:** i) Question paper consists of Part A, Part B.

ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.

iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A**(25 Marks)**

- 1.a) What is distributed database system? [2]
- b) Explain briefly about peer to peer systems. [3]
- c) Define semi join operation. [2]
- d) Write the benefits of normalization. [3]
- e) Define dead lock. [2]
- f) Write the ACID properties of transaction. [3]
- g) Define mutual consistency. [2]
- h) List out communication failures in distributed databases. [3]
- i) Define object and class. [2]
- j) What is object migration? [3]

PART – B**(50 Marks)**

- 2.a) Explain the design issues in Distributed data bases.
 - b) Discuss about vertical fragmentation techniques. [5+5]
- OR**
- 3.a) Explain allocation methods in distributed data bases.
 - b) Write the steps for distributed data base design and briefly discuss. [5+5]
- 4.a) Explain query processing in distributed DBMS with an example.
 - b) What do you mean by the term localization of distributed data? Explain. [5+5]
- OR**
- 5.a) Explain distributed query optimization algorithm.
 - b) Give a detailed note on query decomposition. [5+5]
- 6.a) Explain flat and nested transaction in detail.
 - b) Describe distributed two-phase locking protocol. [5+5]
- OR**
- 7.a) Discuss about optimization concurrency control algorithm.
 - b) Explain deadlock prevention techniques. [5+5]

- 8.a) Explain the components of distributed reliability protocols.
b) Describe three phase commit protocol for distributed reliability. [5+5]

OR

- 9.a) Discuss eager centralized replication protocols.
b) Compare Mutual Consistency versus Transaction Consistency. [5+5]

- 10.a) Explain vertical class partitioning.
b) Describe object query processing architecture. [5+5]

OR

- 11.a) Give a detailed note pointer swizzling.
b) Explain Transactions Management in Object DBMSs. [5+5]

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